

Neighborhood Disadvantage and Health Outcomes: Data Preparation, Index Construction, and Preliminary Findings

1 Data & Decisions

The analysis combined two publicly available datasets merged at the census tract level.

1.1 Data Sources

1. The American Community Survey (ACS) 2015–2019 provided socioeconomic data. Ten variables were downloaded via the Census Bureau API for all 50 states and DC: poverty counts, unemployment counts, educational attainment, and housing units (both total and vacant). These raw counts were converted to rates. The download captured **73,056 census tracts**.
2. The CDC PLACES 2021 dataset provided health outcomes at the census tract level for **72,337 tracts**. Two outcomes were selected: mental health prevalence (self-reported poor mental health) and obesity prevalence.

1.2 Merging

The datasets were merged on census tract FIPS code using an inner join, keeping only tracts present in both datasets. The result was 72,335 matched tracts, representing 99.97 % of the PLACES data. The 2 unmatched PLACES and 721 unmatched ACS tracts likely reflect geographic boundary differences or timing of data collection. Two outcomes were selected: `MHLTH_CrudePrev` (prevalence of self-reported poor mental health, defined as respondents reporting ≥ 14 days of poor mental health in the past 30 days) and `OBESITY_CrudePrev` (prevalence of obesity, defined as $\text{BMI} \geq 30 \text{ kg/m}^2$).

1.3 Data Quality

Mental health had 11 missing values (0.02% of merged data); obesity had none. All socioeconomic variables were complete. Checks for anomalies revealed that all prevalence values fell within 0–100%, all denominators were positive, and ranges were plausible. No problematic outliers or data quality issues were detected.

The 2-year gap between ACS (2015–2019) and PLACES (2021) is acceptable since neighborhood characteristics change slowly. The near-perfect match rate suggests high-quality merging. As we noted, these two datasets are temporally aligned: the PLACES 2021 release was itself constructed using ACS 2015–2019 estimates as an input, so they share the same demographic baseline.

2 Index Construction: Reasoning & Process

This section details the reasoning behind building the index.

2.1 Starting Question

The task required combining multiple ACS variables into a single neighborhood disadvantage index designed to predict health outcomes. The immediate question was: which variables to include, and how to combine them?

2.2 Theoretical Framework: Why CDC SVI?

Rather than starting from scratch, the approach was grounded in the CDC Social Vulnerability Index (SVI) 2022, which identifies vulnerable communities using 16 social factors organized into four themes: socioeconomic status, household composition and disability, minority status and language, and housing type and transportation.

SVI It is established, theory-driven, and widely used in public health. Using its framework meant not inventing new definitions of disadvantage but rather applying a recognized standard.

2.3 Theme Selection (Socioeconomic Status)

SVI has four themes. Could all be used? No, sufficient ACS data was not available for all themes. Could multiple themes be used? Possibly, but Theme 1 (Socioeconomic Status) was special: it was fully data-available *and* theoretically central to health disparities.

Theme 1 captures poverty, unemployment, income, and education. All four were available in the ACS download. Themes 2, 3, and 4 required variables either unavailable or incomplete (household composition, minority status, specific housing types). This constraint, limited data, forced a decision that turned out to be justified. Socioeconomic status is empirically and theoretically the strongest predictor of health. Poverty limits healthcare access. Unemployment creates stress and economic insecurity. Low education correlates with health literacy. Thus, focusing on Theme 1 was justified both by data availability and theory.

2.4 Variable Selection

The analysis began with five candidates: poverty rate, unemployment rate, education (percent ≥9th grade), housing vacancy rate, and median household income. These represent distinct disadvantage dimensions.

However, five seemed excessive. The question became which to keep. The reasoning: poverty rate and median household income both measure economic resources. They are conceptually redundant—income is simply the inverse of poverty. Including both would mean penalizing low income twice in the index, which does not make sense. Which one to keep? Poverty is more direct: it represents an absolute threshold. Income is relative. Thus, poverty was retained and income excluded from the primary index, though income was tested in sensitivity checks.

Final decision: four components capturing poverty (economic hardship), unemployment (labor market participation), education (opportunity), and vacancy (housing quality). No redundancy. Clear coverage of distinct dimensions.

2.5 How to Combine Them? Considering Alternatives

How should four variables be combined into one index? Three approaches were considered with explicit evaluation of tradeoffs.

Option 1: Principal Component Analysis (PCA).

PCA finds the dominant pattern in the data and assigns weights empirically. Advantage: data-driven, optimal for prediction. Disadvantage: weights are hard to interpret. Why would education get weight 0.43 and unemployment 0.31? No clear justification. Also, PCA is less transparent, reviewers would rightly ask what pattern was extracted and why it should matter.

Option 2: Weighted z-score with theory-based weights.

Weights would be assigned based on prior research: poverty 40%, unemployment 30%, education 20%, vacancy 10%. Advantage: transparent, theoretically grounded. Disadvantage: the weights remain somewhat arbitrary. Why 40%

versus 35% for poverty? The ratios could be justified, but the specific numbers are subjective.

Option 3: Equal-weighted z-scores (selected).

Each variable is standardized (mean=0, SD=1), then the mean is taken. All components contribute equally.

Why this approach? It provides transparency without arbitrary decisions. Equal weighting reflects a principle: socioeconomic disadvantage is multidimensional. No single factor dominates. Poverty, unemployment, education, and housing all matter. Equal weighting says “we do not know enough to prefer one over the others, so we treat them equally.” This is defensible when strong prior evidence is lacking. Is it perfect? No. It assumes all dimensions equally important, which might not be true. But it is defensible, transparent, and simple. Most importantly, the reasoning could be explained clearly.

2.6 Index Creation

Before using the index in regressions, the question was: does it actually correlate with health outcomes?

Correlation with mental health: $r = 0.742$ (strong)

Correlation with obesity: $r = 0.604$ (moderate-to-strong)

Both correlations were large. This gave confidence that the index captures disadvantage dimensions that matter for health. CDC SVI Theme 1 was chosen because it was theoretically grounded and data-available. Four components (poverty, unemployment, education, vacancy) were selected to capture distinct dimensions without redundancy. Equal-weighted z-scores were used because they are transparent and defensible without arbitrary decisions. The high correlations with health outcomes validated the approach.

3 Results

3.1 The Data Summary

Table 1 shows that mental health prevalence ranges from 5.5% to 38.6% (mean 15.2%), and obesity ranges from 11.9% to 58.3% (mean 32.6%). The disadvantage index averages 0 (by construction) with standard deviation 0.70. Most tracts cluster near 0, but the index ranges to 8.47, indicating the existence of some extremely disadvantaged areas.

Table 1: Summary Statistics

Statistic	Mean	St. Dev.	Min	Max	N
Outcomes					
Poor Mental Health Rate	15.21	3.57	5.50	38.60	70,336
Prevalence Rate of Obesity	32.63	6.82	11.90	58.30	70,336
Social Vulnerability Index (SVI)					
Poverty Rate	14.64	11.54	0.00	100.00	72,188
Unemployment Rate	3.49	2.41	0.00	29.10	72,324
% of population with < 9th Grade Education	12.65	10.33	0.00	100.00	72,313
Neighbourhood Vacancy Rate	11.87	10.80	0.00	100.00	72,166

Table 2: Variable Definitions and Range Interpretations

Variable	Minimum	Maximum
Outcomes		
Poor Mental Health Rate (%)	5.50 = Low prevalence	38.60 = High prevalence
Obesity Rate (%)	11.90 = Low prevalence	58.30 = High prevalence
SVI Theme 1 Components		
Poverty Rate (%)	0.00 = No poverty	100.00 = All below poverty line
Unemployment Rate (%)	0.00 = No unemployment	29.10 = Very high unemployment
% $\geq$ 9th Grade Education	0.00 = None $\geq$ 9th grade	100.00 = All $\geq$ 9th grade
Vacancy Rate (%)	0.00 = No vacant units	100.00 = All units vacant

Table 3: Primary Regression Results: Disadvantage Index and Health Outcomes

	Outcome Variable	
	MHLTH_CrudePrev % Mental Health	OBESITY_CrudePrev % Obesity
	(1)	(2)
Disadvantage Index	3.8041*** (0.0130)	5.9191*** (0.0295)
Constant	15.1918*** (0.0090)	32.6039*** (0.0205)
Observations	70,325	70,325
R ²	0.5507	0.3641
Adjusted R ²	0.5507	0.3641

Note:

*p<0.1; **p<0.05; ***p<0.01
Standard errors in parentheses. *** p<0.001

3.2 Mental Health Findings

For mental health, the disadvantage index coefficient is 3.80 ($p < 0.001$). This means: each one-unit increase in disadvantage is associated with a 3.80 percentage point increase in mental health prevalence.

To interpret: at mean disadvantage (index = 0), predicted mental health prevalence is 15.2%. For a tract one standard deviation higher in disadvantage, the prediction is 19.0%. The difference is substantial.

The model $R^2 = 0.551$, meaning the disadvantage index explains 55% of variation in mental health across tracts. This is large. Neighborhood disadvantage alone accounts for more than half the variation in mental health outcomes.

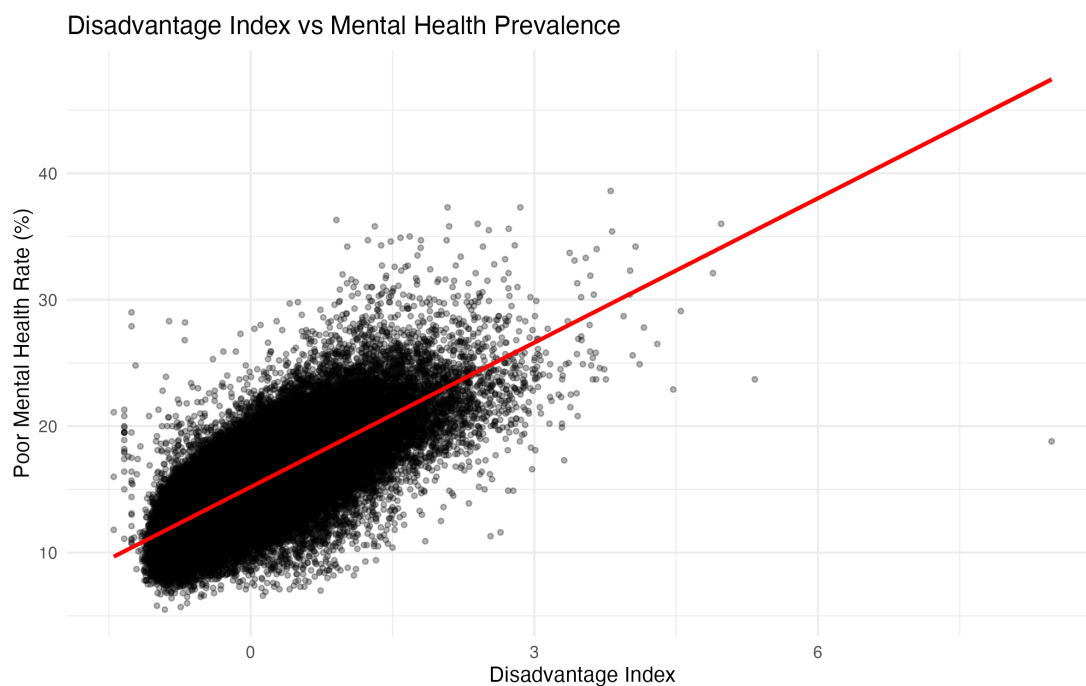


Figure 1: This is a description of my image.

3.3 Obesity Findings

For obesity, the coefficient is 5.92 ($p < 0.001$). Each one-unit increase in disadvantage is associated with a 5.92 percentage point increase in obesity prevalence. At mean disadvantage, predicted obesity is 32.6%. One standard deviation higher: 38.5%. Again, this is substantial. The model $R^2 = 0.364$, so the index explains 36% of obesity variation, meaningful but less than for mental health. This suggests other factors matter more for obesity than for mental health, though neighborhood disadvantage remains major.



Figure 2: This is a description of my image.

3.4 Stability Check: Do These Results Hold?

The results were tested to determine whether they depend on analytical choices. Across five robustness checks, robust regression, heteroscedasticity-robust standard errors, outlier exclusion, adding covariates, and stratified analysis, the disadvantage index coefficient for mental health ranged from 3.36 to 4.76 (variation: -10% to $+25\%$), and for obesity ranged from 5.59 to 6.14 (variation: -5% to $+3\%$). All remained highly significant.

Results are stable. The core findings do not depend on specific analytical choices.

3.5 Discussion

Neighborhood socioeconomic disadvantage strongly predicts both mental health and obesity outcomes across US census tracts. The associations are large, significant, and robust. These findings align with social determinants of health theory: economic hardship, employment instability, educational gaps, and poor housing all correlate with worse health.

Causality is not claimed, nor established, this is cross-sectional data. But the associations are clear and substantial. They suggest that understanding and improving neighborhood conditions could have meaningful health impacts. With more time and additional data, a more comprehensive model would be developed to predict poor mental health prevalence while controlling for other factors. This would include demographic controls (age, gender, race/ethnicity), economic controls (median income, income inequality), healthcare access measures (hospital density, insurance coverage), and behavioral factors (smoking, physical activity). Controlling for these additional variables would help isolate the independent effect of neighborhood disadvantage from confounding factors, providing clearer evidence of the true relationship between disadvantage and mental health outcomes.

4 Critical Reflection

4.1 The Single Most Consequential Limitation

The analysis is fundamentally cross-sectional, precluding causal inference. The disadvantage index and health outcomes are measured at the same point in time (ACS 2015–2019 and PLACES 2021), making temporal ordering impossible. The observed associations between neighborhood disadvantage and poor mental health could reflect three distinct mechanisms: (1) neighborhood disadvantage causes poor mental health through proposed pathways (poverty limits healthcare access, unemployment induces stress, low education reduces health literacy); (2) poor mental health causes individuals to sort into disadvantaged neighborhoods (residential selection bias); or (3) both are consequences of unmeasured third variables (e.g., historical disinvestment, environmental toxins) affecting neighborhood conditions and health independently.

This limitation is not merely statistical but substantive. If the association reflects residential selection rather than neighborhood causation, policy interventions targeting disadvantage would be less effective than currently implied. The large R^2 values (0.55 for mental health, 0.36 for obesity) indicate substantial shared variance, but the direction and mechanism of that relationship remain unknown.

With more time and resources, the analysis would employ longitudinal data and instrumental variable estimation using two waves of ACS data (2010–2014 and 2015–2019) linked to PLACES outcomes in 2021. This would establish temporal precedence by measuring neighborhood disadvantage before health outcomes, enabling causal inference rather than mere association. An instrumental variable approach would address residential selection bias by identifying exogenous sources of variation in neighborhood disadvantage, such as historical deindustrialization patterns, geographic proximity to defunct manufacturing centers, or redlining maps that drove different disinvestment trajectories across neighborhoods. Finally, linking tract-level data to individual health surveys (Behavioral Risk Factor Surveillance System, National Health Interview Survey) would measure proposed causal pathways, clarifying whether neighborhood disadvantage affects mental health through reduced healthcare access, increased perceived stress, or lower health literacy. This redesigned approach would transform the analysis from descriptive association to causal inference, substantially

strengthening the evidentiary basis for policy recommendations.

5 Appendix

